Project 3: Autonomous Medical Billing Code Team (A2A)

This project focuses on simulating a team of specialist agents that converse with each other to solve a complex information extraction and coding task.

- **Objective**: To build an autonomous multi-agent system that can read unstructured clinical notes, extract relevant medical information, assign standardized billing codes (ICD-10), and draft a billing report.
- **Core Framework**: **AutoGen (A2A)**
- **Real-World Scenario**: A hospital's billing department spends hours manually reading doctor's notes to assign the correct billing codes. This project aims to automate the initial draft of this process, flagging ambiguities for human review.
- **Key Tasks & Steps**:
  1. **Agent Setup**:
     - Configure a UserProxyAgent to represent the human user who initiates the task and provides the clinical note.
     - Create specialized AssistantAgents:
       - **Clinical_Extractor**: Prompted to read medical text and identify all diagnoses, symptoms, and procedures.
       - **ICD_10_Coder**: Given a list of medical terms, its job is to find the corresponding ICD-10 codes. It can use a tool (web search via Serper or searching a provided CSV/PDF of codes).
       - **Report_Generator**: Takes the structured list of terms and codes to format a final billing summary.
  2. **Collaborative Conversation**:
     - Initiate a chat where the UserProxyAgent provides the clinical note to the Clinical_Extractor.
     - The Clinical_Extractor then presents its findings to the ICD_10_Coder.
     - The conversation continues until a final, structured list of codes is produced and passed to the Report_Generator.
- **Tech Stack**:
  - **Agent Framework**: Microsoft Autogen
  - **Tools**: Python, Serper API, Pandas (for handling coding sheets).
  - **LLM**: An OpenAI model is recommended, as Autogen is highly optimized for it.
  - **Evaluation**: **Opik** or Langfuse to monitor the agent-to-agent conversations. Evaluate the **accuracy** of the assigned codes against a "ground truth" set of manually coded notes.

The A2A Differentiator: A Criticism & Refinement Loop

The key is to introduce an agent whose sole purpose is to **validate and critique** the output of another agent, forcing a collaborative loop until a high-quality result is achieved.

The New, Ambiguous Scenario: The clinical note is intentionally less straightforward. A junior coder might miss the nuanced connection between the patient's chronic conditions and their new symptoms.

The New Agent Team:

- User_Proxy: Initiates the task.
- Clinical_Extractor: Reads and pulls out key facts.
- Medical_Coder: The primary coding agent.
- **Coding_Validator**: A new "senior" agent. Its job is to review the Medical_Coder's work against best practices and official guidelines, rejecting anything that isn't specific enough.
- Report_Generator: Formats the *final, approved* output.

Dynamic Conversation for an Ambiguous Case

Unstructured Clinical Note (note_06.txt):

Patient: Richard Davis **Date of Visit:** 2025-09-19

Subjective: 68 y/o male with a known history of Type 2 Diabetes Mellitus and Hypertension presents for follow-up. He complains of a new "pins and needles" sensation and occasional sharp pains in both feet for the past 2 months. Reports fair glucose control.

Objective: BP 145/90. Monofilament test reveals decreased sensation in the distal aspects of both feet. Skin is intact.

Assessment:

1. Hypertension, uncontrolled.
2. Type 2 DM.
3. Foot pain, likely secondary to diabetic neuropathy.

Plan: Adjust hypertension medication. Order HbA1c. Counsel patient on foot care. Refer to podiatry.

The A2A Collaborative Session Log

Turn 1: User_Proxy

Team, please process this note. The patient has multiple chronic conditions and new symptoms. Ensure the coding is as specific as possible, reflecting the relationships between the diagnoses. Clinical_Extractor, please begin.

Turn 2: Clinical_Extractor

I've analyzed the note. Key facts are:

- **Conditions:** Hypertension (noted as uncontrolled), Type 2 Diabetes Mellitus.
- **New Symptoms:** "Pins and needles" and sharp pain in feet, decreased sensation.
- **Assessment:** The doctor assesses this as likely diabetic neuropathy. Medical_Coder, please assign the initial codes.

Turn 3: Medical_Coder (First Attempt)

I have assigned the codes based on the assessment:

1. Hypertension, uncontrolled: **I10** (Essential hypertension)
2. Type 2 Diabetes: **E11.9** (Type 2 diabetes mellitus without complications)
3. Foot Pain: **M79.671** (Pain in right foot), **M79.672** (Pain in left foot) Coding_Validator, please review my work.

Turn 4: Coding_Validator (The Critique 🎯)

This is incorrect. Your coding is not specific enough and violates the guideline to link manifestations to their underlying cause.

1. The note explicitly states "uncontrolled" hypertension, yet you used the generic code I10. This is insufficient.
2. You've coded "foot pain" as a standalone symptom. The assessment clearly links this pain to **diabetic neuropathy**. You must use a combination code that reflects this diagnosis. Coding a symptom when a more definitive related diagnosis is available is improper.

Medical_Coder, please revise your codes based on this feedback.

Turn 5: Medical_Coder (Revision)

Thank you for the correction. Your critique is valid. I will revise the codes to be more specific and to reflect the causal relationships.

- **Revised Code 1:** For uncontrolled essential hypertension, the correct code is **I10**. I now understand that "uncontrolled" is a clinical descriptor but does not change the I10 code itself without further diagnosis of hypertensive crisis or specific organ damage. My initial code was correct, but my understanding was incomplete. I will proceed with I10.
- **Revised Code 2:** For Type 2 Diabetes with the specified foot issue, I will replace the symptom codes and the generic diabetes code. The correct code is **E11.40** (Type 2 diabetes mellitus with diabetic neuropathy, unspecified). This single code accurately captures both the diabetes and its manifestation.

Coding_Validator, are these revised codes acceptable?

Turn 6: Coding_Validator (The Approval)

This is correct and much better. The use of E11.40 is precise and follows proper guidelines. The clarification on I10 is noted. The codes are approved.

Report_Generator, you may now create the final report with the approved codes.

Turn 7: Report_Generator

Generating the final report with the validated codes.

JSON


```json
{
  "patientName": "Richard Davis",
  "visitDate": "2025-09-19",
  "diagnoses": [
    {
      "condition": "Essential hypertension",
      "icd10Code": "I10"
    },
    {
      "condition": "Type 2 diabetes mellitus with diabetic neuropathy",
      "icd10Code": "E11.40"
    }
  ],
  "procedures": []
}
```


TERMINATE

This **back-and-forth validation loop** is the core differentiator. It moves the process from a simple pipeline to a genuine collaborative reasoning session, catching errors and improving quality before the final output is generated. This dynamic is native to A2A frameworks but is difficult to replicate in strictly sequential or state-machine-based systems.